

Input Ontology (IO)

Score: 4/5 — Minor gaps | Confidence: high

Benchmark: MRBENCH | Context: Indian Metropolitan Grade 1–8 Mathematics Teachers (Delhi & Mumbai)

Definition: Whether the benchmark’s test case categories cover the query types expected in deployment.

Justification

The user explicitly accepted curriculum-agnostic math errors as sufficient and confirmed the eight pedagogical dimensions are necessary and sufficient (elicitation Q1, Q3), reducing concern at the taxonomy level. The benchmark covers two difficulty levels (elementary and middle-school) [Q143, Q144], aligning with Grade 1–8 scope. The taxonomy is curriculum-agnostic by construction [Q117, Q118] and was validated against learning-science principles with a pilot confirming necessity and sufficiency [Q129, Q131, Q132]. Minor concerns: the taxonomy reflects a Socratic/answer-withholding pedagogical model [Q17, Q23] that the deployment context flags as partially conflicting with Indian direct-correction norms, but this is more an output-rubric concern (handled in OO/OC) than an input-category gap. No Indian curriculum-specific subcategories (e.g., NCERT topics, lakh/crore conventions) are present, but the user explicitly stated these are not required.

Strengths

- Eight-dimension taxonomy was iteratively validated and confirmed necessary and sufficient by pilot annotators [Q129, Q131, Q132]
- Curriculum-agnostic design [Q117] aligns with the user’s explicit acceptance criterion
- Coverage of both elementary and middle-school difficulty levels matches Grade 1–8 scope [Q143, Q144]
- Task is narrowly and clearly defined (mistake remediation in math) [Q12, Q13], reducing construct-irrelevant variance

Checklist

Checklist item definitions:

ID	Assessment Question
IO-1	Identify test case categories required for regional deployment
IO-2	Check if taxonomy omits regionally relevant categories
IO-3	Check if taxonomy includes irrelevant categories
IO-4	Document category gaps harming content validity

Checklist responses:

IO-1: Test case categories required for the deployment are mistake-remediation dialogues at Grade 1–8 math difficulty; the user accepted curriculum-agnostic errors as sufficient (elicitation Q1). The benchmark covers elementary [Q44, Q45] and middle-school [Q48, Q49] difficulty levels.

IO-2: No regionally-relevant categories are documented as omitted at the taxonomy level; the user confirmed the eight dimensions are necessary and sufficient (elicitation Q3). NCERT-specific topics and Indian conventions (e.g., lakhs/crores) are absent but explicitly deemed unnecessary by the user.

IO-3: No taxonomy categories are flagged as irrelevant; all eight dimensions [Q1, Q15] are accepted as applicable to the deployment.

IO-4: No category-level gaps that would harm content validity are identified given the user’s curriculum-agnostic acceptance criterion. Sub-national board variation (CBSE/ICSE/State Board/IB) is a flagged gap but operates at content level rather than taxonomy level.

Information Gaps

- Whether sub-national curriculum variation (CBSE vs. ICSE vs. Maharashtra State Board vs. IB) affects category-level coverage of mistake types relevant to specific teacher subgroups

Requires Expert Verification

- Confirmation from a broader Indian teacher pool (beyond the elicited user) that no additional pedagogical dimensions specific to Indian classrooms (e.g., exam-readiness framing as a distinct dimension) are needed